Syllabus

Math 398: Advanced Research Investigations

Course Information

Term: Fall 2026

Instructor: Isaac Quintanilla Salinas

Contact: isaac.qs@csuci.edu

Office Location: Marin 2326

Office Hours: MW 1-3 PM

Lecture: Tuesday and Thursday 4:30-5:45 PM in Gateway 3550

Course Description

Explore an interdisciplinary mathematical, statistical, or computational research question in independent groups. With faculty mentoring, students develop their own research plans drawing on multiple disciplines and the multiple approaches to research. Students will disseminate results through a research paper and presentations on campus. In addition, students will be encouraged to identify and apply to relevant summer research programs, internships, and scholarships.

Learning Outcomes

1. Evaluate, organize and search various data sets and databases.
2. Apply statistical methods and software to analyze various data.
3. Discuss, visualize and describe various statistical properties of data sets.
4. Apply various data searches and make decisions based on the statistical models.
5. Independently build models and apply appropriate methods to particular data analysis projects to answer a research question.
6. Present the results in visual, oral and written form.
7. Assess and critique in writing various aspects of data analysis and differences in models and results.

Recommended Textbooks

- An Introduction to Statistical Learning in Python (HIGHLY RECOMMENDED)
  - Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor
  - Avalable to Download for free here
- The StatQuest Illustrated Guide to Machine Learning (OPTIONAL)
  - Josh Starmer

Required Software

For this course, we will use several different statistical programs to analyze data and construct machine learning models. **All software is available for free.** In class, we will download and setup your computing environment on your laptop with the following tools:

Programming Language

- **Python** is a general programming langauge that is available to download here.
  - Recommend using tools to create python environments such as uv

Choose One IDE

- **Positron (Recommended)** provides free and open source tools for your data analysis in R and/or Python. You may download positron here.
- **VS Code** provides tools for software development as well as data analysis. You may download VS Code here.
- **VS Codium** is the freely-licensed version of VS Code. You may download VS Codium here.

Course Grading

Category	Percentage
Video Assignments	20%
Weekly Reports	20%
In-class Assignments	20%
Final Report	20%
Final Poster Presentation	20%

At the end of the quarter, course grades will be assigned according to the following scale:

Percentagae	Grade
90 - 100	A
80 - <90	B
70 - <80	C
60 - <70	D
<60	F

Course Assignments

Course Project

Working in a group of up to 2 students, you will construct a data science project by fitting a machine learning model to any data of your interest. Your group will have free range in answering any question related to the project so long as the group can get data. Your couse project will lead to a final report and poster presentation. **Your Poster Presentation will occur on Tuesday 12/8/2026 from 4-6 PM in Gateway Hall.**

Weekly Reports

Weekly reports are progress reports submitted to the instructor to keep track of your progress in completeing the course project. Each week, you will submit what you accomplished in the previous week, any challenges you faced, how you overcame those challenges, and your goal for next week (SMART goal). These will be weekly assignments. Three progress reports will be dropped at the end of the semester. Progress Reports are due every Friday at 11:59 PM. There will be no make ups for progress reports.

Video Assignments

Videos are used to teach machine learning concepts related to the course. Students are expected to watch at least one video a week. The videos are implemented using PlayPosit (WeVideo). The 3 lowest video assignments will be dropped. Video assignments will be due every Monday at 11:59 PM.

In-Class Assignments

In-class assignments are coding assignments where you will progress fitting machine learning models learned during the week. You are expected to complete the assignments every week and submit them every Sunday at 11:59 PM.

Class Schedule

The following outline may be subject to change. Any changes will be announced in class.

Week	Topic	Reading
1 (8/24)	Intro to Course/Project/Computing	2.1 - 2.2
2 (8/31)	Introduction to Machine Learning	3.1 - 3.2
3 (9/7)	Linear Regression	4.1-4.2,4.4
4 (9/14)	Classification	4.3,4.5
5 (9/21)	Logistic Regression	6.2
6 (9/28)	Regularization	5.1,5.2
7 (10/5)	Resampling Methods	7.1-7.4
8 (10/12)	Non-Linear Models	7.5-7.7
9 (10/19)	Generalized Additive Models	8.1
10 (10/26)	Tree-based Methods	8.2
11 (11/2)	Tree-based Methods	9.1-9.4
12 (11/9)	Support Vector Machines	11.1-11.4
13 (11/16)	Survival Analysis	11.5
14 (11/23)	Survival Analysis/**Holiday**	12.1-12.3
15 (11/30)	Unsupervised Learning	
16 (12/7)	Final Presentation	

Generative Artificial Intelligence Policy

The use of generative artificial intelligence (AI) in an ethical manner is permitted for this course.

Permitted Uses

You may use AI for:

- Obtain clarification
- Brainstorming ideas, examples, outlines, and strategies
- Generating questions for practice or exploration
- Identifying keywords or phrasing to match professional goals

Prohibited Uses

You may not:

- Submit AI-generated work
- Use AI to complete assignments, quizzes, exams, or other assessments meant to reflect your own work

Any AI-generated work will receive a 0 in the class. Severe cases will be reported to Academic Misconduct.

You may not upload any course material to any AI platforms such as, but not limited to, ChatGPT, Claude, Github Copilot, Meta AI, or Google Gemini. Exceptions are allowed for DASS-approved services.

University Policies

Syllabus Policies and Assistance

CSUCI's Syllabus Policies and Assistance Website provides important details about academic policies, campus expectations, and student support services that are all highly applicable to your success as a student both in and outside of the classroom. Ensure that you review this site on a regular basis to stay informed about the policies and resources that support your success, as campus resources or policies may change semester to semester.

Academic Honesty

Conduct yourself with honesty and integrity. Do not submit others' work as your own. Foassignments and quizzes that allow you to work with a group, only put your name on what the group submits if you genuinely contributed to the work. Work completely independently on exams, using only the materials that are indicated as allowed. Failure to observe academic honesty results in substantial penalties that can include failing the course.

CSUCI Basic Need

Please use the link to the Basic Needs Program on the Syllabus Policies and Assistance website for information on emergency food, housing accommodations, toiletries, and connections to critical resources.

CSUCI Disability Statement

If you are a student with a disability requesting reasonable accommodations in this course, you need to contact Disability Accommodations and Support Services (DASS) located on the second floor of Arroyo Hall, via email accommodations@csuci.edu or call 805-437-3331. All requests for reasonable accommodations require registration with DASS in advance of need. Faculty, students and DASS will work together regarding classroom accommodations. You are encouraged to discuss approved.

Disruption

1. **If I Am Out:** I will communicate via email and will hold classes asynchronously.
2. **If You Are Out:** Contact me as soon as possible to talk about your options. Reasonable accommodations will be provided for a brief absence. With proper documentation, extended accommodations will be provided.